



THE UNIVERSITY
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Biotechnology and
Biological Sciences
Research Council



THE ROYAL
SOCIETY

Theory: Introduction

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My Quantitative Genetics Journey

- First learned Quant. Gen. in plant breeding classes
 - Relied heavily on assumptions and simplifications
- Truly learned Quant. Gen. while developing AlphaSimR
 - Assumptions and simplification don't work
- Applying theory to optimize breeding programs
 - AlphaSimR is one tool in the toolbox

Areas of Focus

- Augment traditional Quant. Gen. training
 - Review some formulas
 - Conceptual introduction to complex concepts
 - Provide AlphaSimR demonstrations
- Use of theory to tune simulations
 - Empirical data should guide simulation
 - Simulation should guide empirical data collection

Historic use of Quant. Gen. in Breeding

- The Breeder's Equation
 - Popularized by Lush
 - See Hill, 2014
 - Uses specific assumptions
 - Can be generally applied
- Optimization of phenotyping
 - Fisher and block designs
 - See R/selectiongain
 - Inbred line development

$$Gain = \frac{ih\sigma_A}{L}$$

i : Selection intensity

h : Accuracy

σ_A : Diversity

L : Generation interval

From Infinitesimal to Finite

- Classic theory uses the infinitesimal model
 - Assumes an infinite number of causal loci
 - Each has an infinitesimal average effect
 - It is not assumed that all effects are additive
 - Usually uses deterministic simulations
- Many finite loci behave similarly to the infinitesimal model
 - Behavior changes with fewer loci (Mendelian vs Polygenic)
 - Typically modeled using stochastic simulation

Why use Stochastic Simulations?

- Doesn't require deterministic formulas
 - Long-term genetic gain
 - Non-random mating
 - Genomic selection
- Can handle greater complexity
 - Model whole breeding programs
 - Test more targeted questions
- Modern computing makes it feasible

Upcoming Topics

- Additive effects
 - Introduce theoretical concepts
- Dominance effects
 - Emphasis on tuning simulations
- Epistatic effects
 - Scratch the surface
- Genetic maps and recombination
- Autopolyploids